CHAPTER 7

# Coordinate geometry: lines, projections and circles

**Chapter path**

**Objectives.** Use a normal vector; calculate a projection; determine or recognise a circle; translate a configuration into equations.
**Prerequisites.** Vector coordinates, scalar product, linear systems, square completion.
Coordinate system always orthonormal.
**Yearly reference point.** 5, 7 and 12 January: 6 hours

## Diagnostic check: six questions, 10 minutes

Answer without looking at your notes. Indicate the questions for which help is needed. This diagnostic test is not graded.

1. $(2;-1)\cdot(1;2)$ ?

2. Does the point (1; 2) satisfy 2 $x-y=0$ ?

3. Solve $x+y=5,\ x-y=1.$

4. Distance from (1; 2) to (4; 6) ?

5. Complete $x^2-4x.$

6. Solve $x^2+y^2$ = 0 in $\mathbb{R}^2.$

**After the diagnostic check.** Go over the flagged concepts with the teacher; redo a question similar, then explain the rule used.

**Oral work and logic.** Present the two conditions characterising a projection. Give a proof
geometric proof and an algebraic proof of a perpendicular bisector.

## Investigate before formalising

From $P(4; 3)$ and $d : y = 2x$, compare the distances to three points of d, then find and prove which one is shorter.

**Organisation.** Individual work 5 min, discussion in pairs 5 min, whole-class sharing 5 min. Keep a record of an attempt, even an unfinished one. A conjecture must then be justified.

## 7.1 Normal vector and Cartesian equation

**Definition and results**

A non-zero normal vector to a line is orthogonal to every direction vector of that Right. $\vec{n} = (a; b) \neq (0; 0)$ is normal and $A(x_A; y_A)$ belongs to d, then $d$ has equation $a(x - x_A) + b(y - y_A) = 0$, i.e. $ax + by + c = 0$. A possible direction vector is ( $-b; a)$.

**Status and justification.** $M \in d$ is equivalent to $\overrightarrow{AM} \cdot \vec{n} = 0$. In an orthonormal coordinate system, expand this product gives the equation. Conversely the set thus defined is the line passing through A orthogonal to n.

**Worked example.** By $A(1; 2)$ and normal $(2; -3)$ : $2(x - 1) - 3(y - 2) = 0$, therefore $2x - 3y + 4 = 0$. The vector (3; 2) directs the line: product $6 - 6 = 0$.

**Watch out**

$(a; b)$ is normal, and in general not a direction vector. ( $a; b)$ must not be zero.

**Immediate application.** Equation of the line through (2; $-1)$ with normal vector (3; 4).

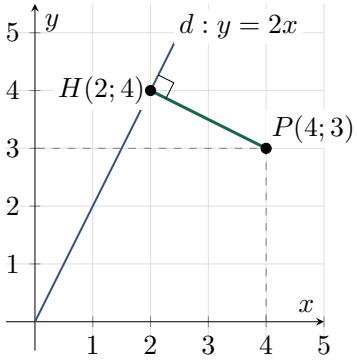
## 7.2 Orthogonal projection and distance

**Definition and results**

The projection H of P onto d is characterised by $H \in d$ and $\overrightarrow{PH}$ collinear with a normal vector of d. For $d : ax + by + c = 0$, write $H = P - \lambda(a; b)$, with $\lambda = \dfrac{ax_P + by_P + c}{a^2 + b^2}$. The distance is $PH = \dfrac{|ax_P + by_P + c|}{\sqrt{a^2 + b^2}}$. The distance formula is a derived complement, not a fluency item isolated.

**Status and justification.** Replace $x_H = x_P - \lambda a$, $y_H = y_P - \lambda b$ in the equation: $ax_P + by_P + c - \lambda(a^2 + b^2) = 0$. The denominator is positive; we obtain H, then $PH = |\lambda| \sqrt{a^2 + b^2}$.

**Worked example.** For $P(3; 4)$ and $d : x + y - 2 = 0$, $\lambda = \dfrac{3 + 4 - 2}{2} = \dfrac{5}{2}$, therefore $H(\dfrac{1}{2}; \dfrac{3}{2})$ and $PH = \dfrac{5}{\sqrt{2}} = \dfrac{5\sqrt{2}}{2}$. Check $x_H + y_H = 2$.




The segment PH is perpendicular to d. The two axes have the same scale; the coordinates of P and H can be read directly.

**Watch out**

A distance is positive or zero: keep the absolute value. Two checks: membership and perpendicularity, confirm the projection.

**Immediate application.** Project $P(4; 1)$ on $d : x = 2$.

## 7.3 Equation of a circle

**Definition and results**

The circle with centre $I(\alpha; \beta)$ and radius $r > 0$ has equation $(x - \alpha)^2 + (y - \beta)^2 = r^2$. A equation $x^2 + y^2 + ux + vy + w = 0$ is handled by completing both squares. If the squared radius square obtained is negative, the set is empty; if it is zero, it is a point.

**Status and justification.** $M$ belongs to the circle if and only if $IM = r$. Both sides being positive or zero, we can square and use the distance formula in an orthonormal coordinate system.

**Worked example.** $x^2 + y^2 - 4x + 6y + 9 = 0$ becomes $(x - 2)^2 + (y + 3)^2 = 4$. Centre $(2; -3)$, radius 2. The signs of the centre's coordinates are opposite to those in the brackets.

**Watch out**

A quadratic equation in x and y is not always a circle: check the coefficients and the real radius.

**Immediate application.** Recognize $x^2 + y^2 + 2x - 8y + 17 = 0$.

## 7.4 Solving a configuration

**Definition and results**

An intersection is found by solving the equations simultaneously. For a circle and a line parallel to an axis, substituting the constant coordinate gives a quadratic equation degree. For a perpendicular bisector, write the equality of the squared distances; the quadratic terms cancel out.

**Status and justification.** The equivalences rely on membership of both sets. The equality $MA = MB$ is equivalent to $MA^2 = MB^2$, since the distances are positive.

**Worked example.** The circle $x^2 + y^2$ = 25 and the line $x$ = 3 meet at (3; 4) and (3; $-4$). The perpendicular bisector of $A$(0; 0) and $B$(4; 2) satisfies $x^2 + y^2 = (x-4)^2 + (y-2)^2$, therefore $2x + y = 5$.

**Watch out**

An intersection is not the union of the solution sets. Check the points in both equations.

**Immediate application.** Intersect ( $x - 1)^2 + (y + 2)^2$ = 9 with $y = 1$.

## Exercise path

Start with exercises 1 and 2. Then choose exercise 3 or 4 without help. Exercises 5 to 7 are a reserve for transfer, investigation or programming: the teacher chooses one. Exercise 8 is revisited later.

### Exercise 7.1 · Constructing a line

The plane is given an orthonormal coordinate system. The line $d$ passes through $A(-1; 3)$ and has the vector $\vec{n}$ = (2; 5) as normal vector.

1. Write the condition for a point to belong $M(x; y)$ à $d$ using a dot product. In deduce a Cartesian equation from $d$.
2. Give a nonzero direction vector of $d$, then the coordinates of a point of $d$ distinct from $A$.
3. Check your point by substituting into the equation.

### Exercise 7.2 · Calculating a projection

In an orthonormal coordinate system, we are given the point $P$(4; 3) and the line $d$ with equation 2 $x - y$ = 0. $H$ the orthogonal projection of $P$ on $d$.

1. Determine the coordinates of $H$ by translating the two conditions: $H$ belongs to $d$ and $\overrightarrow{PH}$ is normal to $d$.
2. Check these two conditions separately with the coordinates obtained.
3. Calculate the distance from $P$ right $d$ in exact form.

## Exercise 7.3 · Recognising a circle

Consider the set of points $M(x; y)$ of the orthonormal plane whose coordinates satisfy

$$x^2 + y^2 - 6x + 4y - 12 = 0.$$

1. Show that this set is a circle and specify its centre and radius.
2. Determine all its intersection points with the line with equation $y = -2$.
3. Interpret geometrically the position of these two points on the circle.

## Exercise 7.4 · Perpendicular bisector

In an orthonormal coordinate system, we are given $A(-1; 1)$ and $B(3; 3)$. We seek an equation of the perpendicular bisector of $[AB]$.

1. Determine this equation using the geometric properties of the perpendicular bisector. Explain the choice of a point and a normal vector.
2. Verify that the midpoint of $[AB]$ satisfies the equation obtained.

## Exercise 7.5 · The best access

A straight road is represented by the line $d$ with equation $y = 2x$ in an orthonormal coordinate system whose unit of length is 1 km. A building is located at the point $P(4; 3)$. We want to construct a straight access road linking the building to a point on the road.

1. Determine the point on the road that makes this access as short as possible. Calculate the length exact value of the access.
2. Justify that your choice is optimal across all points of the road, comparing distances using a right-angled triangle. A single observation on a drawing is not enough.

## Exercise 7.6 · Parameter and Set

For a real parameter $k$, we define $\mathcal{E}_k$ as the set of points $M(x; y)$ whose coordinates satisfy

$$x^2 + y^2 - 2x + 4y + k = 0.$$

1. Determine the type of $\mathcal{E}_k$ depending on the value of $k$. Organise the answer into disjoint cases.
2. When this set is a circle, specify its centre and radius; in other cases, describe exactly the set obtained.

## Exercise 7.7 · A parabola as a locus

**Optional extension.** In an orthonormal coordinate system, fix the point $F(0; 2)$. We look for points $M(x; y)$ whose distance to $F$ is equal to the distance to the x-axis.

1. Express the square of each of the two distances in terms of $x$ and $y$.
2. Deduce an equation of the required set. Check the converse to justify that this equation describes exactly all the required points.

### Exercise 7.8 · Delayed review

In an orthonormal coordinate system, $\mathcal{C}$ is the circle of diameter $[AB]$, with $A(0;0)$ and $B(4;0)$. Consider the line $d$ with equation $x = 5$.

1. Determine whether $\mathcal{C}$ and $d$ have a common point, by solving an equation.

2. Confirm the answer by comparing the distance from the centre of the circle to the line with the radius of the circle.

**Independent work.** Do only the task assigned at the end of the session, in the scheme of work Date: 10·20 min maximum. The exercises depend on the concepts already studied; some questions are therefore postponed. Revisit exercise 8 one week after the chapter (10 min). If the planned time is exceeded, note the difficulty and bring your attempts; do not use up the whole reserve.

## To be done in the CQFD textbook

These exercises are done in your textbook. The scheme of work gives the session; they replace the personal work planned that day (10 to 20 minutes in total).

**CQFD, Exercise 18 p. 279, questions a, b, c and d.** Read a non-zero normal vector for each equation, without solving for y.

**CQFD, Exercise 21 p. 279, questions a and b.** Write the condition of orthogonality to the normal, then check that the given point lies on the line.

## MCQ 1: understand and choose

Five teaching questions with a single answer. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official format of the EAM.

1. A normal vector to 2 $x - 3y + 1 = 0$ is

A. $(3; 2)$
B. $(2; -3)$
C. $(2; 3)$
D. $(1; 0)$

2. A direction vector of $x + 2y = 0$ is

A. $(-2; 1)$
B. $(1; 2)$
C. $(2; 1)$
D. $(0; 2)$

3. Centre of $x + 2)^2 + (y - 3)^2 = 4$ :

A. $(2; 3)$
B. $(-2; -3)$
C. $(3; -2)$
D. $(-2; 3)$

4. Radius of $x^2 + y^2 = 9$ :

A. $9$
B. $81$
C. $3$
D. $\sqrt{3}$

5. $x^2 + y^2 = -1$ describes

A. a circle
B. the empty set
C. the origin
D. a line

## MCQ 2: understand and choose

Five teaching questions with a single answer. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official format of the EAM.

1. Projection of (3; 4) onto the x-axis:

A. $(0; 4)$
B. $(0; 0)$
C. $(3; 0)$
D. $(4; 3)$

2. Distance from (1; 5) to $y = 2$ :

A. $3$
B. $5$
C. $1$

D. $\sqrt{10}$

3. To define $ax + by + c = 0$ to be a line, we need

A. $a \neq 0$ only

B. $b \neq 0$ only

C. $c \neq 0$

D. $(a; b) \neq (0; 0)$

4. $x^2 + y^2 = 0$ describes

A. A circle of radius 1

B. the single point O

C. a line

D. the plane

5. Circle $x^2 + y^2 = 4$ and the line $x = 2$ : number of common points

A. $0$

B. $2$

C. $1$

D. Infinite

## True or false? Justify.

1. The orthogonal projection of P onto a line is unique.

2. Every equation $x^2 + y^2 + c = 0$ describes a circle of positive radius.

3. A line can meet a circle at three distinct points.

**Personal review**

I give a non-zero normal vector; I check the projection; I complete both squares; I distinguish circle, point and empty set.
For each objective, indicate: mastered without help; achieved with help; needs more work. Write an example and a mistake to avoid.

# Solutions and teaching notes, chapter 7

**Teacher use.** The following answers concern the tasks not solved in the lesson. Do first make the approach explicit; provide graduated help before the solution. The durations are estimates to be calibrated in class.

## Diagnostic and remediation

1. $0$. *Action:* Go back to the orthogonality criterion.

2. Yes: $2 \quad -2=0$. *Action:* Substitute x and y simultaneously.

3. $x=3,\ y=2$. *Action:* Add, then substitute.

4. $5$. *Action:* Calculate the vector and then its norm.

5. $(x-2)^2-4$. *Action:* Expand to check.

6. The only pair (0; 0). *Action:* Use the sum of two positive squares.

## Investigation

Trials suggest H(2;4), but Pythagoras' theorem in PHM proves minimality. The projection is calculated with the normal (2;-1).

## Direct applications

1. $3(x-2)+4(y+1)=0$, i.e. $3 \quad x+4y-2=0$.

2. $H(2;\ 1)$ and distance 2; the normal is horizontal.

3. $(x+1)^2+(y-4)^2=0$ : the single point $(\ -1;\ 4)$, not a circle of strictly positive radius.

4. $(x-1)^2=0$, so only the point (1; 1); the line is tangent to the circle.

## Exercises: methods and answers

**Exercise 7.1 · Constructing a line.** $2(x+1)+5(y-3)=0$, i.e. $2 \quad x+5y-13=0$. Direction vector $(5;-2)$, therefore $B(4;\ 1)$ is on the line; check $8+5 \quad -13=0$.

**Exercise 7.2 · Calculating a projection.** Normal vector $(2;\ -1)$ ; $\lambda=\dfrac{8-3}{5}=1$. $H=(4;3)-(2;-1)=(2;4)$. $2(2)-4=0$ and $\overrightarrow{PH}=(-2;\ 1)$ is a normal vector; distance $\sqrt{5}$.

**Exercise 7.3 · Recognising a circle.** $(x-3)^2+(y+2)^2=25$; centre $(3; \quad -2)$, radius 5. For $y=-2,\ x=3\pm5$ : $(-2;-2)$ and $(8;-2)$.

**Exercise 7.4 · Perpendicular bisector.** Midpoint I(1;2), normal $\overrightarrow{AB}=(4;2)$. $4(x-1)+2(y-2)=0$, i.e. $2x+y-4=0$. I satisfies $2+2 \quad -4=0$.

**Exercise 7.5 · The best access.** The projection H(2;4) minimises PH, with length $\sqrt{5}$ km. For every M on the road, triangle PHM is right-angled at H and $PM^2=PH^2+HM^2\geq PH^2$. Equality for M=H.

**Exercise 7.6 · Parameter and set.** $(x-1)^2+(y+2)^2=5-k$. If $k<5$, circle with centre $(1;-2)$ and radius $\sqrt{5-k}$; if $k=5$, a point; if $k>5$, empty.

**Exercise 7.7 · A parabola as a locus.** $MF^2=x^2+(y-2)^2$, squared distance to the axis $y^2$. Equality: $x^2-4y+4=0$, or $y=\frac{x^2}{4}+1$. Positive distances allow the converse to be used.

**Exercise 7.8 · Delayed review.** Centre (2; 0), radius 2 ; $(x-2)^2+y^2=4$. For x=5, $y^2=-5$, impossible. Interpretation: the line is at distance 3 from the centre, greater than the radius.

**CQFD, Exercise 18 p. 279, questions a, b, c and d.** Possible answers are $(-1;2)$, $(3;-6)$, $(\frac{1}{4};-1)$ and (1; 0). Any nonzero multiple works. For $x=4$, write $x-4=0$.

**CQFD, Exercise 21 p. 279, questions a and b.** a : $x-(y-2)=0$, i.e. $x-y+2=0$. b : $4(x+1)+6(y-3)=0$, i.e. $x+3y-7=0$. Substituting the coordinates of the points confirms membership.

## MCQ: answers and reasons

**MCQ 1.**

1. B. Coefficients of x and y.

2. A. Zero product with the normal vector.

3. D. Read the coordinates with opposite signs.

4. C. Positive square root of 9.

5. B. A sum of squares is nonnegative.

**MCQ 2.**

1. C. Same x-coordinate, zero y-coordinate.

2. A. Absolute vertical distance.

3. D. Nonzero normal vector.

4. B. Both squares must be zero.

5. C. $y^2=0$.

## True or false

1. True: through P there is a unique line perpendicular to the given line; if P lies on the line, H=P.

2. False: if c=0, a single point; if c>0, the empty set.

3. False: substituting a parametrisation of a line into the equation of a circle gives an equation quadratic with strictly positive leading coefficient, hence at most two roots.

**Differentiated teaching.** If a pupil is stuck on a technical point, suggest only the first step (domain, factor or reference sample space), then remove the help. For an independent student, ask for a second method and the handling of a boundary case. Orally, allow an initial rephrasing in the language mastered, then require a precise mathematical sentence in French.